

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I G. Ramani(192312279) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student:

Question 1 — Case Study 1: Smart Medical Diagnosis System

Q1(a) Propositional Logic Representation

Symbols used:

- F = Patient has Fever
- C = Patient has Cough
- H = Patient has Rash
- B = Patient has Breathlessness
- L = Possible Flu
- M = Possible Measles
- P = Possible Pneumonia

Rules:

- R1: $F \wedge C \rightarrow L$
- R2: $F \wedge H \rightarrow M$
- R3: $C \wedge B \rightarrow P$

Facts (Patient A): F = True, C = True, B = True, H = False / Unknown

Q1(b) Forward Chaining

Step	Rule Checked	Condition	Result
1	R1: $F \wedge C \rightarrow L$	$F = T, C = T \rightarrow$ satisfied	Fire $\rightarrow$ L (Flu) = True
2	R2: $F \wedge H \rightarrow M$	H unknown/False $\rightarrow$ not satisfied	Not fired
3	R3: $C \wedge B \rightarrow P$	$C = T, B = T \rightarrow$ satisfied	Fire $\rightarrow$ P (Pneumonia) = True

Derived facts: L (Possible Flu), P (Possible Pneumonia)

Q1(c) Backward Chaining — Goal: P (Pneumonia)

Goal: P

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+-- needs: C AND B    (Rule III)
  +-- C -> check KB -> Fact: C = True  [OK]
  +-- B -> check KB -> Fact: B = True  [OK]
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Both subgoals satisfied $\Rightarrow P = \text{True}$ (GOAL PROVED)

Question 2 — Case Study 2: Autonomous Traffic Management Agent

Q2(a) Unification with Rule 1

Rule 1: $\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

- Unify $\text{Vehicle}(x)$ with $\text{Vehicle}(\text{Ambulance}) \rightarrow \theta_1 = \{x/\text{Ambulance}\}$
- Unify $\text{EmergencyType}(x)$ with $\text{EmergencyType}(\text{Ambulance})$ under $\theta_1 \rightarrow$ consistent
- MGU: $\theta = \{x/\text{Ambulance}\}$
- Apply to conclusion $\rightarrow \text{ClearPath}(\text{Ambulance})$ derived

Q2(b) Resolution Proof — Prove Proceed(CarA)

CNF of the rules:

- 1. $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$
- 2. $\neg \text{ClearPath}(y) \vee \text{GreenSignal}(y)$
- 3. $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Facts (unit clauses): $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

Negated goal: $\neg \text{Proceed}(\text{CarA})$

Resolution steps:

- ```
1) (Clause 1)+Vehicle(Ambulance) {x/Ambulance}
 -> ¬EmergencyType(Ambulance) v ClearPath(Ambulance)
2) + EmergencyType(Ambulance) -> ClearPath(Ambulance)
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3) (Clause 2){y/Ambulance} + ClearPath(Ambulance)  $\rightarrow$  GreenSignal(Ambulance)  
 4) (Clause 3){x/CarA,y/Ambulance} + Vehicle(CarA)  
 $\rightarrow \neg \text{Behind}(\text{CarA}, \text{Ambulance}) \vee \neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$   
 5) + Behind(CarA,Ambulance)  $\rightarrow \neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$   
 6) + GreenSignal(Ambulance)  $\rightarrow \text{Proceed}(\text{CarA})$   
 7) +  $\neg \text{Proceed}(\text{CarA})$  (negated goal)  $\rightarrow []$  EMPTY CLAUSE

**Conclusion: Proceed(CarA) is proved TRUE.**

### Q2(c) Sufficiency of the KB for Two Simultaneous Emergency Vehicles

**The current KB is NOT sufficient. Gaps identified:**

- No predicate for priority/ordering between two competing ambulances at the same junction.
- No rule to resolve conflict when two ClearPath requests intersect (e.g., perpendicular roads).
- Needed additions: Priority(x), Conflict(x,y), and a rule such as:  $\text{Vehicle}(x) \wedge \text{Vehicle}(y) \wedge \text{Conflict}(x,y) \wedge \text{Priority}(x) > \text{Priority}(y) \rightarrow \text{ClearPath}(x) \wedge \neg \text{ClearPath}(y)$ .
- Also needs timestamp/arrival-order facts to break ties when priorities are equal.

### Question 3 — Case Study 3: Agricultural Expert Reasoning System

#### Q3(a) CNF Conversion

Symbols: D = SoilDry, I = IrrigationNeeded, W = CropWheat, A = ApplyDripMethod, R = CropAtRisk

- P1:  $D \rightarrow I \Rightarrow \neg D \vee I$
- P2:  $I \wedge W \rightarrow A \Rightarrow \neg(I \wedge W) \vee A \Rightarrow \neg I \vee \neg W \vee A$
- P3:  $\neg A \wedge W \rightarrow R \Rightarrow \neg(\neg A \wedge W) \vee R \Rightarrow A \vee \neg W \vee R$
- Facts: D, W (already unit clauses)

#### Q3(b) Resolution Refutation — Prove ApplyDripMethod (A)

Clauses: (1)  $\neg D \vee I$  (2)  $\neg I \vee \neg W \vee A$  (3)  $A \vee \neg W \vee R$  (4) D (5) W (6)  $\neg A$  (negated goal)

D (4) +  $\neg D \vee I$  (1)  $\rightarrow I$   
 I +  $\neg I \vee \neg W \vee A$  (2)  $\rightarrow \neg W \vee A$   
 $\neg W \vee A$  + W (5)  $\rightarrow A$   
 A +  $\neg A$  (6, negated goal)  $\rightarrow []$  (empty clause)

**Conclusion: ApplyDripMethod (A) is proved TRUE.**

#### Q3(c) If Fact 'SoilDry' (D) is Removed

- P1 ( $D \rightarrow I$ ) can no longer fire, so I cannot be derived.
- P2 ( $I \wedge W \rightarrow A$ ) cannot fire without I, so A cannot be proven true.
- Under strict classical logic: A is undetermined (neither proved true nor false), so P3's antecedent  $\neg A \wedge W$  cannot be confirmed either  $\rightarrow R$  (CropAtRisk) remains undetermined, not proven.

- Under the Closed World Assumption (common in expert systems — 'if it cannot be proven true, treat as false'): failure to derive  $A \Rightarrow$  assume  $\neg A$ . Then  $\neg A \wedge W(\text{True}) \rightarrow R$  fires, so  $\text{CropAtRisk} = \text{True}$  still follows — but only via this non-monotonic reasoning pattern, not classical entailment.

#### Question 4 — Case Study 4: Wumpus World Knowledge Agent

##### Q4(a) Belief State via Modus Ponens

Percepts:  $\text{Stench}[1,2] = \text{True}$ ,  $\text{Breeze}[1,1] = \text{True}$ ,  $\text{Glitter}[2,2] = \text{True}$

| Rule | Premise                          | Modus Applied         | Ponens                                        | Derived Fact                               |
|------|----------------------------------|-----------------------|-----------------------------------------------|--------------------------------------------|
| 1    | $\text{Stench}[1,2] = \text{T}$  | $\text{Stench}[1,2]$  | $\rightarrow$<br>$\text{WumpusAdjacent}[1,2]$ | $\text{WumpusAdjacent}[1,2] = \text{True}$ |
| 2    | $\text{Breeze}[1,1] = \text{T}$  | $\text{Breeze}[1,1]$  | $\rightarrow$<br>$\text{PitAdjacent}[1,1]$    | $\text{PitAdjacent}[1,1] = \text{True}$    |
| 3    | $\text{Glitter}[2,2] = \text{T}$ | $\text{Glitter}[2,2]$ | $\rightarrow$<br>$\text{Gold}[2,2]$           | $\text{Gold}[2,2] = \text{True}$           |

**Derived conclusions:**  $\text{WumpusAdjacent}[1,2]$ ,  $\text{PitAdjacent}[1,1]$ ,  $\text{Gold}[2,2]$

##### Q4(b) Forward Chaining — Locating the Wumpus and the Pit

- Neighbours of  $[1,2]$  are  $\{[1,1], [1,3], [2,2]\}$ . The agent is alive at  $[1,1]$  with no stench recorded there, so  $\text{Wumpus} \neq [1,1] \rightarrow \text{Wumpus} \in \{[1,3], [2,2]\}$ .
- Neighbours of  $[1,1]$  (corner cell) are  $\{[1,2], [2,1]\}$ . Since  $\text{Breeze}[1,1]$  is true  $\rightarrow \text{Pit} \in \{[1,2], [2,1]\}$ .
- Applying Rule 4 ( $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$ ): cell  $[1,1]$  itself is Safe (the agent survived entering it); no other cell can yet be certified Safe since the candidate sets above remain ambiguous.

##### Q4(c) Is the Path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ Safe?

- $[1,2]$ : Cannot rule out a Pit — it is one of the two Pit candidates identified in Q4(b). Rule 4 cannot certify it Safe  $\rightarrow$  risk of Pit.
- $[2,2]$ : It is one of the two Wumpus candidates identified in Q4(b)  $\rightarrow$  risk of Wumpus, despite containing the Gold.

**Conclusion:** The path is NOT logically proven safe. Neither  $[1,2]$  nor  $[2,2]$  satisfies  $\neg \text{Wumpus} \wedge \neg \text{Pit}$ , so Rule 4 cannot assert Safe for either cell. The agent should first visit  $[2,1]$  or  $[1,3]$  to gather additional Breeze/Stench percepts and narrow down the Pit/Wumpus locations via further forward chaining before committing to this path.